

FACTORS & MULTIPLES

Grade 5

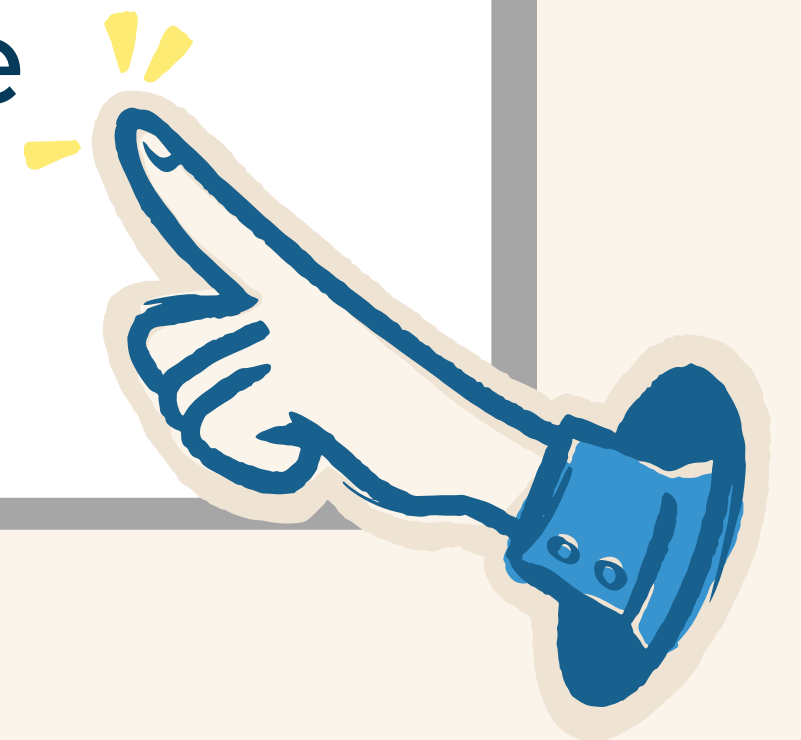




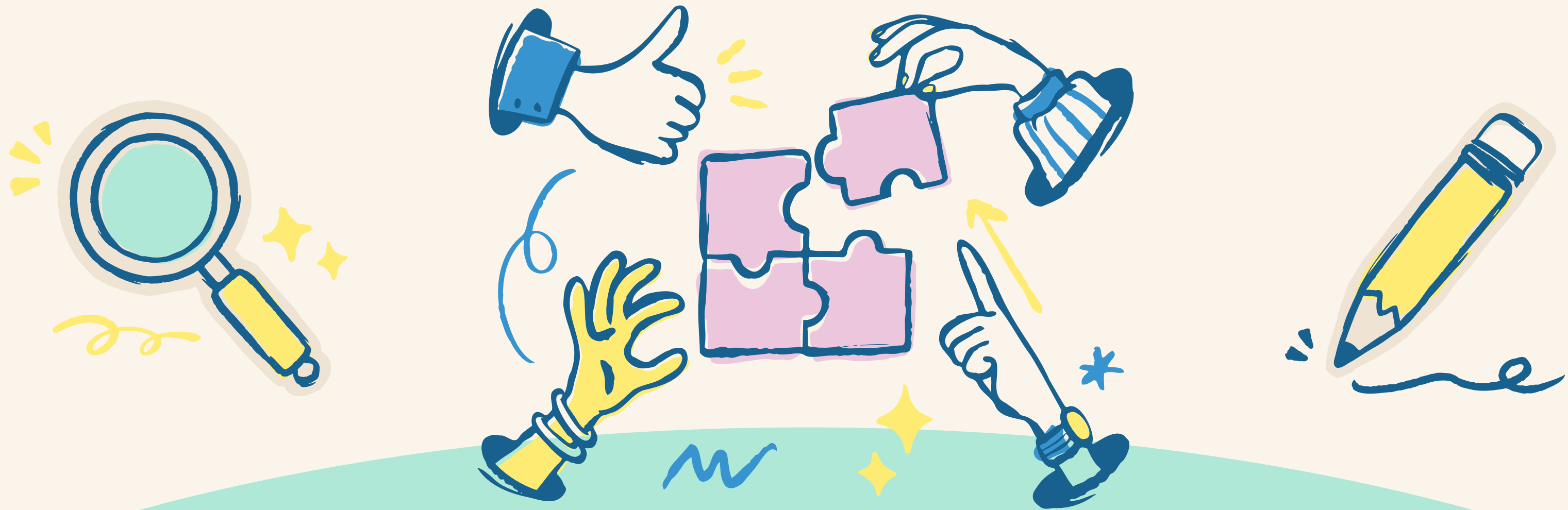
WHAT IS A FACTOR?

Factors are numbers we can multiply together to get another number.

A number is a factor if it can divide evenly into the other number.



LET'S SEE SOME EXAMPLES!



WHAT ARE THE FACTORS OF 6?

$$1 \times 6 = 6$$

$$2 \times 3 = 6$$

So, the factors of 6
are 1, 2, 3, and 6.



Think about all the
whole numbers you
can multiply to get 6.



WHAT ARE THE FACTORS OF 12?

Remember:
A factor of a number
divides the number
evenly, with no
remainder.

$$1 \times 12 = 12$$

$$3 \times 4 = 12$$

$$2 \times 6 = 12$$

Can we multiply
something by 5 to
get 12?

6 was already used!
So we are done.

So, the factors of 12 are
1, 2, 3, 4, 6, and 12.

WHAT ARE THE FACTORS OF 16?

$$1 \times 16 = 16$$

$$2 \times 8 = 16$$

$$4 \times 4 = 12$$

Can we multiply something
by 3 to get 16?

4 was used twice,
so that means we
stop there!

So, the factors of 16 are
1, 2, 4, 8, and 16.



Ask yourself:
“Can I multiply this
number and something
else to get 16?”

HOW DO I FIND ALL THE FACTORS OF 24?

Start with 1.

You can multiply 1×24 . So, 1 and 24 are both factors of 24.

Put 1 at the beginning of your factor list and 24 at the end.



1

24

HOW DO I FIND ALL THE FACTORS OF 24?

Now try 2.

You can multiply 2×12 to get 24. So, 2 and 12 are both factors of 24.

Add them to our list.

1

2

12

24



HOW DO I FIND ALL THE FACTORS OF 24?

Now try 3.

You can multiply 3×8 to get 24. So, 3 and 4 are both factors of 24.

Add them to our list.



1 2 3 8 12 24

HOW DO I FIND ALL THE FACTORS OF 24?

Now try 4.

You can multiply 4×6 to get 24. So, 4 and 6 are both factors of 24.

Add them to our list.



1 2 3 4 6 8 12 24

HOW DO I FIND ALL THE FACTORS OF 24?

Now try 5.

We can't divide 24 by 5 evenly, so it is not a factor of 24. Let's move on to 6.

6 is already on our list, so that means our list is complete!



1 2 3 4 6 8 12 24

HOW DO I FIND ALL THE FACTORS OF 24?



The factors of 24 are
1, 2, 3, 4, 6, 8, 12, and 24



NOW IT'S YOUR TURN!

1. Find the factors of 32
2. Which number has more factors: 12 or 15?
3. Find the factor pairs of 36
4. A class is making teams. There are 30 students. What are all the ways they can divide evenly into teams?



Practice Questions



- 1 Find the factors of 32
- 2 Which number has more factors: 12 or 15?
- 3 Find the factor pairs of 36
- 4 A class is making teams. There are 30 students. What are all the ways they can divide evenly into teams?



WHAT IS A MULTIPLE?

A multiple of an number is what you have get when you multiply that number by 1, 2, 3, and so on.

Example:

$$3 \times 1 = 3$$

$$3 \times 2 = 6$$

$$3 \times 3 = 9$$

So the multiples of 3 are: 3, 6, 9...





IMPORTANT FACTS

- The first multiple of every numbers is the number **itself**.
- Every number is a **multiple of itself**.
- Multiples get **bigger** and **bigger** as you go.
- You can get multiples by **skip counting**.



Practice Questions



- 1 Write the first 6 multiples of 7
- 2 Is 21 a multiple of 5?
- 3 Which numbers from 1 to 30 are multiples of both 3 and 6?
- 4 What is the smallest number that is a multiple of both 4 and 5?
- 5 Find all the multiples of 9 that are less than 75